

same colour also for $ii' = e_{m-1}$. Continuing in this way, we finally arrive at an induced copy of G^0 in G such that, for each pair (i, i') , all the edges between V_i^0 and $V_{i'}^0$ have the same colour. As shown earlier, this G^0 contains a monochromatic induced copy H_j of H . $\square$

The Ramsey graph G for H built in the proof of Theorem 9.3.1 is sparse in the sense that it contains no complete graphs bigger than it has to in order to contain H : G and H have the same clique number $\omega(G) = \omega(H)$ (Exercise 23). This property, which comes for free with this proof of the theorem, is non-trivial even if we do not ask for induced embeddings: the existence of a graph G that contains a monochromatic copy of H , induced or not, on any 2-colouring of its edges is neither obvious nor does it follow easily from Ramsey's original theorem.

An even stronger sparsity requirement on G is to ask that it should contain no shorter cycles than H does, that $g(G) = g(H)$. As a small indication of how ambitious this requirement on G is, notice that the statement of $\chi(G) > r$ can be rephrased as a Ramsey-like property of G : for any vertex-colouring with at most r colours, G contains a 'monochromatic' copy of $H = K^2$, an edge joining two like-coloured vertices. But as we noted in Chapter 5.2, graphs of simultaneously large chromatic number and girth are difficult to construct.

However, Ramsey graphs of large girth do exist:

Theorem 9.3.4. (Reiher & Rödl 2023)

Every graph H has a Ramsey graph G that satisfies $g(G) = g(H)$ unless H is a forest, in which case $g(G)$ can be made arbitrarily large.

If H is a tree but not a star, the aim of $g(G) = g(H)$ ($= \infty$) cannot be realised: if $g(G) = \infty$ then G is a forest, so its edges can be 2-coloured alternately so that every path of length 3 receives two colours.

9.4 Ramsey properties and connectivity

According to Ramsey's theorem, every large enough graph G has a very dense or a very sparse induced subgraph of given order, a K^r or $\overline{K}^r$. If we assume that G is connected, we can say a little more:

Proposition 9.4.1. *For every integer $r \geq 1$ there is an $n \in \mathbb{N}$ such that every connected graph of order at least n contains K^r , $K_{1,r}$ or P^r as an induced subgraph.*

(1.3.3) *Proof.* Let $d+1$ be the Ramsey number of r , let $n \geq \frac{d}{d-2}(d-1)^r$, and let G be a graph of order at least n . If G has a vertex v of degree at least

$d+1$ then, by Theorem 9.1.1 and the choice of d , either $N(v)$ induces a K^r in G or $\{v\} \cup N(v)$ induces a $K_{1,r}$. On the other hand, if $\Delta(G) \leq d$, then by Proposition 1.3.3 G has radius $> r$, and hence contains two vertices at a distance $\geq r$. Any shortest path in G between these two vertices contains a P^r . $\square$

In principle, we could now look for a similar set of ‘unavoidable’ k -connected subgraphs for any given connectivity k . To keep these ‘unavoidable sets’ small, it helps to relax the containment relation from ‘induced subgraph’ for $k = 1$ (as above) to ‘topological minor’ for $k = 2$, and on to ‘minor’ for $k = 3$ and $k = 4$. For larger k , no similar results are known.

Proposition 9.4.2. *For every integer $r \geq 2$ there is an $n \in \mathbb{N}$ such that every 2-connected graph of order at least n contains C^r or $K_{2,r}$ as a topological minor.*

Proof. Let d be the n associated with r in Proposition 9.4.1, and let G be a 2-connected graph with at least $\frac{d}{d-2}(d-1)^r$ vertices. By Proposition 1.3.3, either G has a vertex of degree $> d$ or $\text{diam}(G) \geq \text{rad}(G) > r$. (1.3.3)

In the latter case let $a, b \in G$ be two vertices at distance $> r$. By Menger’s theorem (3.3.6), G contains two independent a – b paths. These form a cycle of length $> r$. (3.3.6)

Assume now that G has a vertex v of degree $> d$. Since G is 2-connected, $G - v$ is connected and thus has a spanning tree; let T be a minimal tree in $G - v$ that contains all the neighbours of v . Then every leaf of T is a neighbour of v . By the choice of d , either T has a vertex of degree $\geq r$ or T contains a path of length $\geq r$, without loss of generality linking two leaves. Together with v , such a path forms a cycle of length $\geq r$. A vertex u of degree $\geq r$ in T can be joined to v by r independent paths through T , to form a $TK_{2,r}$. $\square$

Theorem 9.4.3. (Oporowski, Oxley & Thomas 1993)

For every integer $r \geq 3$ there is an $n \in \mathbb{N}$ such that every 3-connected graph of order at least n contains a wheel of order r or a $K_{3,r}$ as a minor.

Let us call a graph of the form $C^n * \overline{K^2}$ ($n \geq 4$) a *double wheel*, the 1-skeleton of a triangulation of the cylinder as in Fig. 9.4.1 a *crown*, and the 1-skeleton of a triangulation of the Möbius strip a *Möbius crown*.

Theorem 9.4.4. (Oporowski, Oxley & Thomas 1993)

For every integer $r \geq 4$ there is an $n \in \mathbb{N}$ such that every 4-connected graph with at least n vertices has a minor of order at least r that is a double wheel, a crown, a Möbius crown, or a $K_{4,r}$.

Note that the graphs listed in Theorems 9.4.3 and 9.4.4 are themselves 3-connected resp. 4-connected, as required.



Fig. 9.4.1. A crown and a Möbius crown

At first glance, the ‘unavoidable’ substructures presented in the four theorems above may seem to be chosen somewhat arbitrarily. In fact, the contrary is true: these sets are smallest possible, and as such unique.

non-trivial
property

$\leq$

$\sim$

To make this precise, call a graph property *non-trivial* if it contains graphs of infinitely many isomorphism types. Given two such properties $\mathcal{P}, \mathcal{P}'$ and an order relation $\leq$ between graphs (such as the subgraph relation $\subseteq$, or the minor relation $\preceq$), write $\mathcal{P} \leq \mathcal{P}'$ if for every $G \in \mathcal{P}$ there is a $G' \in \mathcal{P}'$ such that $G \leq G'$. If $\mathcal{P} \leq \mathcal{P}'$ as well as $\mathcal{P} \geq \mathcal{P}'$, call $\mathcal{P}$ and $\mathcal{P}'$ *equivalent* and write $\mathcal{P} \sim \mathcal{P}'$. For example, if $\leq$ is the subgraph relation, $\mathcal{P}$ is the class of all paths, $\mathcal{P}'$ is the class of paths of even length, and $\mathcal{S}$ is the class of all subdivisions of stars, then $\mathcal{P} \sim \mathcal{P}' \leq \mathcal{S} \not\leq \mathcal{P}$.

Kuratowski
set

Given a non-trivial graph property $\mathcal{G}$, call a finite set $\{\mathcal{P}_1, \dots, \mathcal{P}_k\}$ of non-trivial graph properties $\mathcal{P}_i \subseteq \mathcal{G}$ a *Kuratowski set* for $\mathcal{G}$ and $\leq$ if the $\mathcal{P}_i$ are incomparable (i.e., $\mathcal{P}_i \not\leq \mathcal{P}_j$ whenever $i \neq j$) and for every non-trivial graph property $\mathcal{P} \subseteq \mathcal{G}$ there is an i such that $\mathcal{P}_i \leq \mathcal{P}$. Such a Kuratowski set $\{\mathcal{P}_1, \dots, \mathcal{P}_k\}$ is unique up to equivalence: if $\{\mathcal{Q}_1, \dots, \mathcal{Q}_\ell\}$ is another Kuratowski set for $\mathcal{G}$ then $\ell = k$ and, with suitable enumeration, $\mathcal{Q}_i \sim \mathcal{P}_i$ for $i = 1, \dots, k$. (Why?)

The essence of our last four theorems can now be stated more comprehensively, as follows. Let us say *k-connectedness* for the class of all *k*-connected finite graphs, and *connectedness* for 1-connectedness.

Theorem 9.4.5.

- (i) The stars and the paths form the (2-element) Kuratowski set for connectedness and the subgraph relation.
- (ii) The cycles and the graphs $K_{2,r}$ ($r \geq 2$) form the (2-element) Kuratowski set for 2-connectedness and the topological minor relation.
- (iii) The wheels and the graphs $K_{3,r}$ ($r \geq 3$) form the (2-element) Kuratowski set for 3-connectedness and the minor relation.
- (iv) The double wheels, the crowns, the Möbius crowns, and the graphs $K_{4,r}$ ($r \geq 4$) form the (4-element) Kuratowski set for 4-connectedness and the minor relation. $\square$

Exercises

- 1.⁻ Determine the Ramsey number $R(3)$.
- 2.⁻ Deduce the case $k = 2$ (but c arbitrary) of Theorem 9.1.3 directly from Theorem 9.1.1.
3. Prove that the graphs not containing some fixed star as an induced subgraph are χ -bounded.
(See Chapter 5.6 for the definition of χ -boundedness.)
4. Let the edges of a complete graph G on $\mathbb{N}$ be coloured with two colours. Show that G contains two monochromatic paths whose vertex sets partition $\mathbb{N}$.
5. An *arithmetic progression* is an increasing sequence of numbers of the form $a, a + d, a + 2d, a + 3d \dots$. *Van der Waerden's theorem* says that no matter how we partition the natural numbers into two classes, one of these classes will contain arbitrarily long arithmetic progressions. Must there even be an infinite arithmetic progression in one of the classes?
6. Can you improve the exponential upper bound on the Ramsey number $R(n)$ for perfect graphs?
7. Show that the n^ϵ -type bound in the Erdős-Hajnal conjecture cannot be lowered substantially, by finding a graph H such that infinitely many graphs G not containing H as an induced subgraph satisfy $\alpha(G) \leq |G|^{1/2}$ as well as $\omega(G) \leq |G|^{1/2}$.
8. Let $\mathcal{G}$ be a class of graphs that is closed under taking induced subgraphs. Assume that $\mathcal{G}$ is χ -bounded, with a polynomial χ -bounding function. Prove the Erdős-Hajnal conjecture for the graphs in $\mathcal{G}$.
- 9.⁺ Construct a graph on $\mathbb{R}$ that has neither a complete nor an edgeless induced subgraph on $|\mathbb{R}| = 2^{\aleph_0}$ vertices. (So Ramsey's theorem does not extend to uncountable sets.)
10. (i)⁻ Use Ramsey's theorem to show that for any $k, \ell \in \mathbb{N}$ there is an $n \in \mathbb{N}$ such that every sequence of n distinct integers has an increasing subsequence of length $k + 1$ or a decreasing subsequence of length $\ell + 1$.
(ii) Prove that $n = k\ell + 1$ has this property but $n = k\ell$ does not.
- 11.⁺ Find a short proof of the Erdős-Pósa theorem (2.3.2) with a recursively defined bound.
(Hint. For the recursion step find two cycles with at most one vertex in common. If Menger's theorem between them does not give you k disjoint cycles, use the size of the separator it offers for the definition of $f(k)$ from $f(k - 1)$.)
- 12.⁺ Prove the edge version of the Erdős-Pósa theorem (2.3.2): there exists a function $g: \mathbb{N} \rightarrow \mathbb{R}$ such that, given $k \in \mathbb{N}$, every graph contains either k edge-disjoint cycles or a set of at most $g(k)$ edges meeting all its cycles.
(Hint. Consider in each component a normal spanning tree.)